

The nonabelian Fourier–Stieltjes idempotent norm gap

Literature-implied answer note for arXiv:math/0405063

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Status. `literature_implied_answer` (full question).

Source question

Ilie and Spronk [1] prove that a nonzero idempotent $u \in B(G)$ has norm one exactly when it is the indicator of an open coset. Immediately after their Theorem 2.1, on journal page 488 (PDF page 9), they recall Saeki’s abelian gap

$$\|u\|_{B(G)} = 1 \quad \text{or} \quad \|u\|_{B(G)} \geq c_0, \quad c_0 = \frac{1 + \sqrt{2}}{2},$$

and write: “We do not know if a similar result holds for nonabelian groups.”

Supporting theorem

Mudge and Pham [2, Theorem 2.2] prove a stronger statement. For every locally compact group G , if a nonzero idempotent $u = \chi_S \in M_{cb}A(G)$ satisfies

$$\|u\|_{cb} < c_0,$$

then S is an open coset. The theorem is on PDF page 3. Their introduction, on PDF page 2, records the contractive inclusion

$$B(G) \subseteq M_{cb}A(G), \quad \|v\|_{cb} \leq \|v\|_{B(G)} \quad (v \in B(G)).$$

Identification

Let $u \in B(G)$ be a nonzero idempotent with $\|u\|_{B(G)} < c_0$. The norm comparison gives $\|u\|_{cb} < c_0$, so Mudge–Pham’s theorem makes u the indicator of an open coset. Ilie–Spronk’s Theorem 2.1(i) then gives $\|u\|_{B(G)} = 1$. Therefore, for every locally compact group, including every nonabelian one,

$$\|u\|_{B(G)} = 1 \quad \text{or} \quad \|u\|_{B(G)} \geq \frac{1 + \sqrt{2}}{2}.$$

The constant is sharp for the class of all locally compact groups because the abelian examples establishing sharpness remain members of that class.

Provenance and scope

This is an agent-identified implication, rather than a new mathematical result. The later theorem is stated for the larger algebra $M_{cb}A(G)$ and therefore answers the source sentence after applying its explicit norm comparison. A bounded search used the exact source sentence, the threshold, the terms “idempotent norm gap” and “nonabelian Fourier–Stieltjes”, and later citing papers; no stronger interpretation of the source question was found. The original question, as stated, has no remaining case.

References

- [1] M. Ilie and N. Spronk, *Completely bounded homomorphisms of the Fourier algebras*, J. Funct. Anal. **225** (2005), 480–499; [arXiv:math/0405063](#).
- [2] J. Mudge and H. L. Pham, *Idempotents with small norms*, J. Funct. Anal. **270** (2016), 4597–4603; [arXiv:1510.03535](#).